

Evolutionary relationship and protein structure analysis of *WRKY* TF in different species. **A.** Phylogenetic tree analysis of WRKY transcription family factors which respond H<sub>2</sub>S signal. **B.** WRKY71 is a classical zinc-finger protein.

1 10 20 30 40 50 60 70 80 90 100 110 120 130

S1MRKY30 MSDDNPFHHDIYHPFFGEPSTYGNQANNTSPHHHHIDYSSYSSHTLCELGHGSMOHY--NTL--SSRFGHNNCS--SSVDDCPQV--TDHHEQCRKSNISVD--NOITP  
S1MRKY71 NGDELRLLYHOP--FQEDTKAGNTHNTHHIDL--PTHTSYETLHGSSSDYH--GKPGFSTLSLKKDQKHQD--VVRNAYVNNNDVVGSESTP  
ALMRKY28 HNSCTRLDLYNYQYSPFSLHEHNHLPSTIMPSS--YGNLPSQNGFNPTSYFTDCLQSPRAYE--SLL--QKTFGLSPSSSEVFNSIDQEP--NRQDYVNIHGNCAR  
ALMRKY78 HSEIKLNDLYNHTSYNNHNNHNLPLPVS--PSGPNAYHNNSSQVGLDPSLKNLSPQ--GAF--PDYGLSPSSOFFPNLSDQENGLYNNHNNSSKHEVYGGDCAT  
O5MRKY16 HSSGGGGGGGGGRHGPYHQHHLGRGEGADYVYSSDSMEFFSQPQGGVIGGGGGGVYVGGAGDHPYSSITDYLQLLDPSLARHLADYCPSSQDTPVKQELS--VDTSHSGDQGVGAGEVGR  
ALMRKY71 MDHVEYHNNTLSEEVHNSLSDCLSSQLVDHNSLKEVFKSPSSYFQVSPSYNNPYNL

Consensus .e.l .l .f .ss .ll .vd .qe .d .q .v.v.

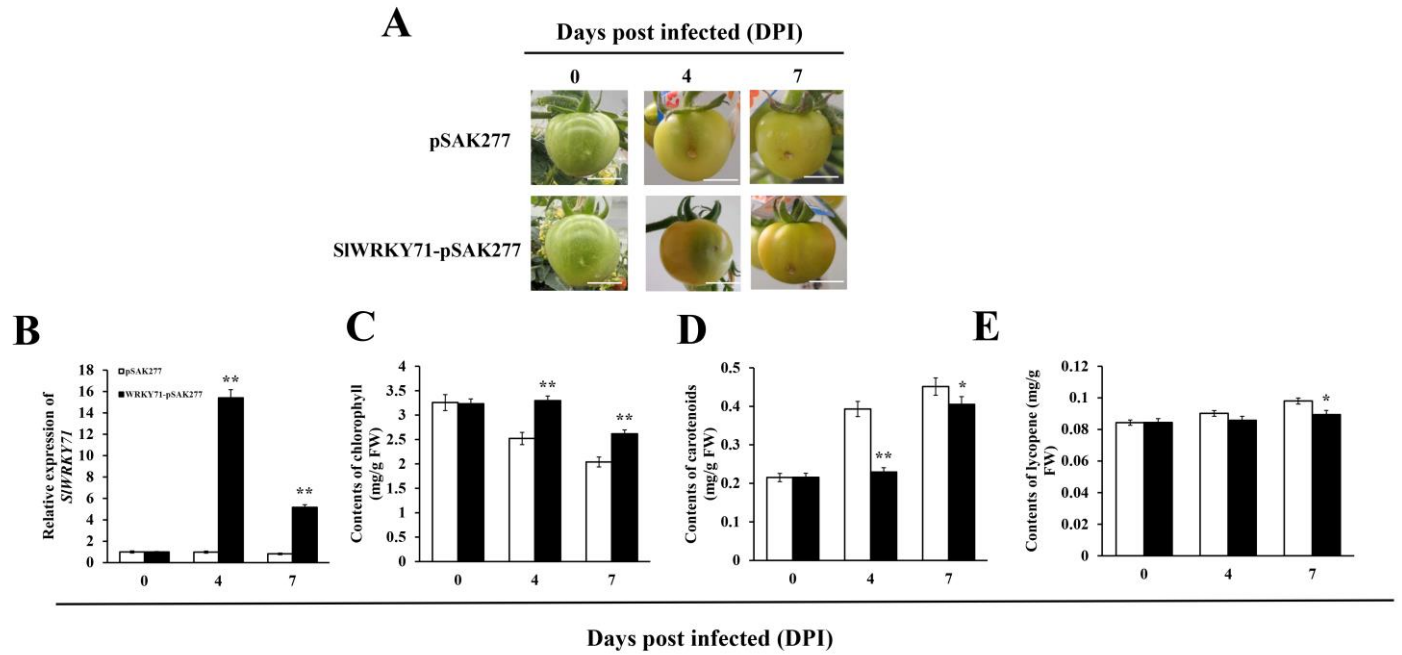
131 140 150 160 170 180 190 200 210 220 230 240 250 260

S1MRKY30 LTPNLSISSNSSEVGGCHEEDSSKITKDDQCELDGGDDDDNDKSKYVGLAKKKGEKKQKEPRFATKSEIQLDLDGGRYARKYQKRVKNPPSPSTYRCTSKCVSKKRVSEYEDPVIITVITTE  
S1MRKY71 VTPNLSISSNS--GHEDNSDKQKQVQKDELTDEGD--AKKESK--AKKGEKKQKPPKFAFTKSEIQLDLDGGRYARKYQKRVKNPPSPSTYRCTSKCVKVKRVSEYEDPVIITVITTE  
ALMRKY28 -ETETVSSSSSEADHPGDEGSKRRKRLV--GEED--ISKVKG--TKKTEVYKQKRPVFATKSEIQLDLDGGRYARKYQKRVKNPPSPSTYRCTSKCNVKKRVSEYEDPVIITVITTE  
ALMRKY78 IKSEVYVSPSSSEADHPGDEGSKRRKRLV--GEED--RSQVKV--TKKKEKK--KEPVSFATKSEIQLDLDGGRYARKYQKRVKNPPSPSTYRCTSKCNVKKRVSEYEDPVIITVITTE  
O5MRKY16 ATNRSKYSSEDEHGGKSRACKQKARKKEEDKQDEGKSPKN--PKCKRKQKQKPFATKSEIQLDLDGGRYARKYQKRVKNPPSPSTYRCTSKCVKVKRVSEYEDPVIITVITTE  
ALMRKY71 LTPNLSISSNEGEPATNKSDDSGDGLL--E--G--SKK--K--KKKgeKkqrpvafATKSeIQLDLDGGRYARKYQKRVKNPPSPSTYRCTSKCVKVKRVSEYEDPVIITVITTE  
Consensus .t.p.s.s.s .h .k .e .d.g.d .Skk .K .KKKgeKkqrpvafATKSeIQLDLDGGRYARKYQKRVKNPPSPSTYRCTSKCVKVKRVSEYEDPVIITVITTE

261 270 280 290 300 310 320 330 340 350 360 370 38882

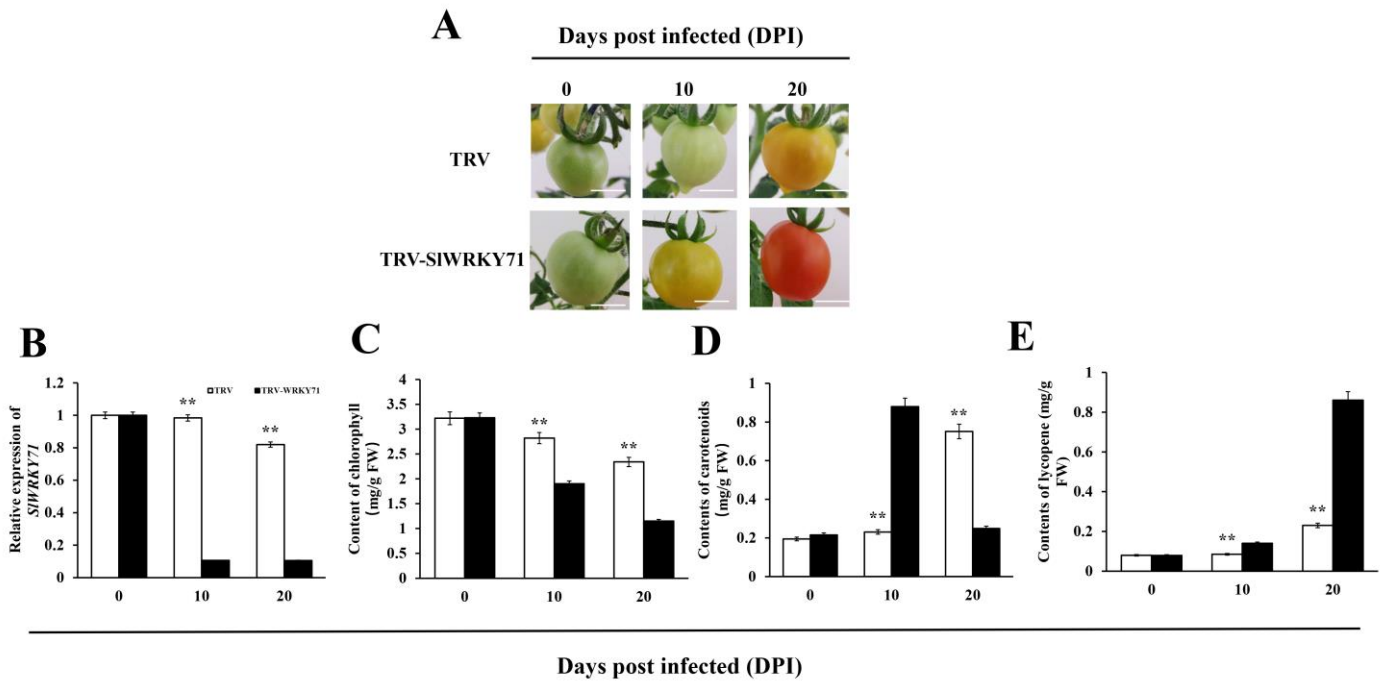
S1MRKY30 QHNHQPATLRGNA--AALLSPSFLSPSQQQLYH--PNEQVGYHNPIINNSFY--NNHYHQQPOLG--PDYGLFQDNYASLNLKCEP  
S1MRKY71 QHNHLPATLRGNAHLPNLPALNPQSPALNPQAFQGEHQZHQSPQVGYHNPVNNNSFY--QHNHNAHQQPOLPH--PHADGLQVYASFLNKP  
ALMRKY28 QHNHPTPLNGSSNAHNSDLNLT--SFADHRTAYNGGSHVHVALDYGAGGSSQVNSPNSHNYVY--QEGTELEIFLSPF--QKQV  
ALMRKY71 QHNHPTPLNGSSNAHNSDLNLT--SFADHRTAYNGGSHVHVALDYGAGGSSQVNSPNSHNYVY--QEGTELEIFLSPF--QKQV  
O5MRKY16 QHNHPTPLNGSGGGGGVGTGGAHNLHPVGHQLPSPSLHPAGFPELNLGLHNNHAPAAAANNSYTFPGVRAASPPPVRAAGGAPRPNHPLLQHNHITDVALQLDFPSTSKNES  
ALMRKY71 KHNHNPSTPLRTGTVAHEHLLVIRGGGGGLHSPFPHHQGLLHMKHSPANYQS--VGLSLSEYHGHTSSTYNNFNQ--PYVDGGLQDTPHSTSKNES  
Consensus qhnhp.PatLRG--A .l .n .g .n .yq .g .gylldmvp.f.k.kep

Supplemental Figure S2.



Functional analysis of *WRKY71* in tomato fruit via the transient overexpression assay. **A**. Phenotype of tomato tissues infected by *WRKY71*-pSAK277 at 7 days. **B**. *WRKY71* expression increased in tissues infected by *WRKY71*. **C-E**. Chlorophyll, carotenoids and lycopene contents in tomato tissues infected by *WRKY71*. The values are the means  $\pm$  SDs of three replicates. Scale bars = 1 cm. Statistical analysis was carried out with Student's *t*-test, and significance was marked with asterisks \* ( $p < 0.05$ ) or \*\* ( $p < 0.01$ ).

Supplemental Figure S3.



Functional analysis of *WRKY71* in tomato fruit via the virus-induced gene silencing (VIGS) assay. **A.** Phenotype of tomato tissues infected by TRV-*WRKY71* at 7 days. **B.** *WRKY71* expression decreased significantly in tissues infected by *WRKY71*-TRV. **C-E** Chlorophyll, carotenoids and lycopene contents in tomato tissues infected by *WRKY71*-TRV. The values are the means  $\pm$  SDs of three replicates. Scale bars = 1 cm. Statistical analysis was carried out with Student's *t*-test, and significance was marked with asterisks \* ( $p < 0.05$ ) or \*\* ( $p < 0.01$ ).

Supplemental Figure S4.

**A**

Sample: 0002\_32022101300629\_(w1)\_[F].ab1

Anchor sequence: AAGTCAAAAATGTCAAGTGA

Degenerate sequence: AGAAAGCGTGTAGAGA(Homozygous mutant)

Allele1: AAGTCAAAAATGTCAAGTGAAAGAAGCGTGTAGAGA

Allele2: AAGTCAAAAATGTCAAGTGAAAGAAGCGTGTAGAGA

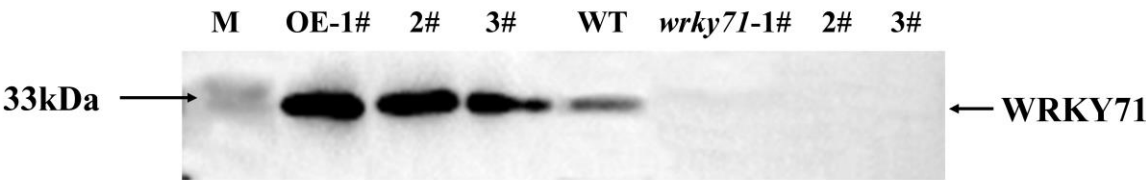
Alignments of Allele1, Allele2, and reference sequence:

Allele1: AAGTCAAAAATGTCAAGTGAAAGAAGCGTGTAGAGAG (insertion)

Allele2: AAGTCAAAAATGTCAAGTGAAAGAAGCGTGTAGAGAG (insertion)

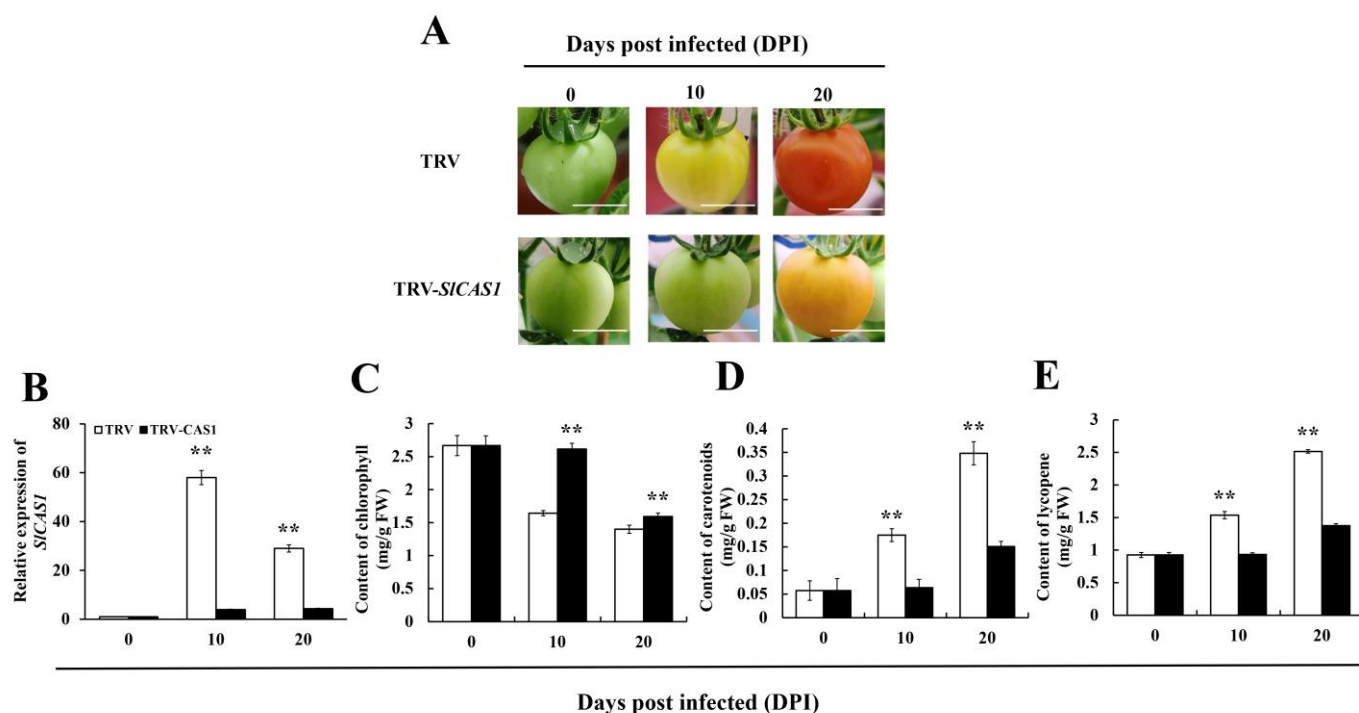
Reference: AAGTCAAAAATGTCAAGTGA-AGAAGCGTGTAGAGAG

**B**



Identification of CRISPR/Cas9-*SIWRKY71* tomato fruit (*wrky71*) and stable *SIWRKY71* transgene tomato (*SIWRKY71*-OE). **A** There was an A was inserted into allele 1 and allele 2 of CRISPR/Cas9-SIWRKY71 tomato fruit. **B** Expression levels of *SIWRKY71* protein in different strains.

Supplemental Figure S5.



Analysis of *CAS1* in tomato fruit via the virus-induced gene silencing (VIGS) assay. **A.** Phenotype of tomato tissues infected by *TRV-CAS1* at 7 days. **B.** *CAS1* expression decreased significantly in tissues infected by *CAS1*-TRV. **C-E** Chlorophyll, carotenoids and lycopene contents in tomato tissues infected by *CAS1*-TRV. The values are the means  $\pm$  SDs of three replicates. Scale bars = 1 cm. Statistical analysis was carried out with Student's *t*-test, and significance was marked with asterisks \* ( $p < 0.05$ ) or \*\* ( $p < 0.01$ ).

## Supplemental Figure S6.

**A**

Sample: 0007\_32020110203634\_(A-1)\_[CAS-1].ab1

Anchor sequence: TTTCATCAATGGCGTCGTTG

Degenerate sequence: TGTGAGGAGAAGATT (Homozygous mutant)

Allele1: TTTCATCAATGGCGTCGTTGTGTGAGGAGAAGATT

Allele2: TTTCATCAATGGCGTCGTTGTGTGAGGAGAAGATT

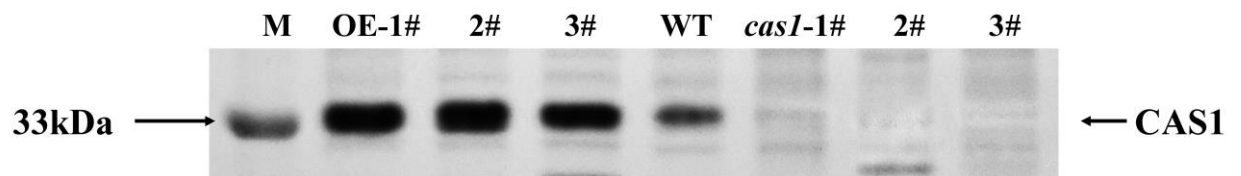
Alignments of Allele1, Allele2, and reference sequence:

Allele1: TTTCATCAATGGCGTCGTTGTGTGAGGAGAAGATTT (insertion)

Allele2: TTTCATCAATGGCGTCGTTGTGTGAGGAGAAGATTT (insertion)

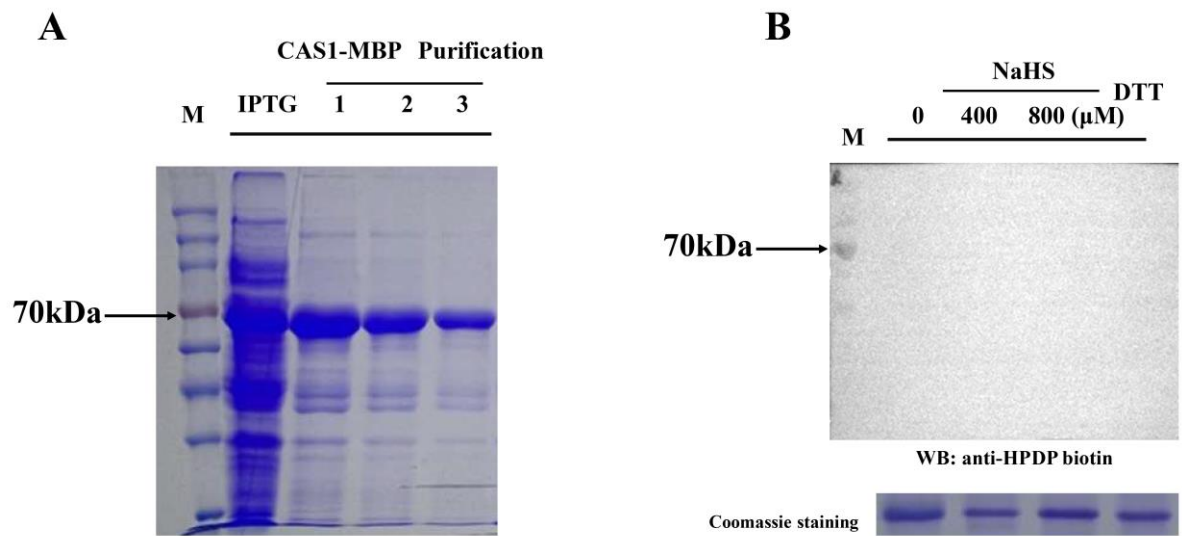
Reference: TTTCATCAATGGCGTCGTTG-GTGAGGAGAAGATTT

**B**



Identification of CRISPR/Cas9-*SlCAS1* tomato fruit (*cas1*) and stable *SlCAS1* transgene tomato (*SlCAS1*-OE). **A** There was an A inserted into allele 1 and allele 2 of CRISPR/Cas9-*SlCAS1* tomato fruit. **B** Expression levels of *SlCAS1* protein in different strains.

## Supplemental Figure S7



Detection of persulfidation signal in SlCAS1. **A.** CAS1 protein was expressed and purified by *Escherichia coli*. **B.** Detection of persulfidation signal in CAS1 protein.

**Supplemental Table S1.** Transcriptome data filtering and quality assessment.

| Sample | Total Reads | Unmapped Reads   | Unique Mapped Reads | Multiple Mapped reads | Mapping Ratio |
|--------|-------------|------------------|---------------------|-----------------------|---------------|
| CK_0d  | 65554754    | 5636961 (8.60%)  | 59336455 (90.51%)   | 581338 (0.89%)        | 91.40%        |
| H2S_1d | 68615108    | 6570183 (9.58%)  | 61415139 (89.51%)   | 629786 (0.92%)        | 90.42%        |
| CK_1d  | 59151404    | 5205437 (8.80%)  | 53408707 (90.29%)   | 537260 (0.91%)        | 91.20%        |
| H2S_3d | 66043502    | 7322124 (11.09%) | 58200846 (88.13%)   | 520532 (0.79%)        | 88.91%        |
| CK_3d  | 62568768    | 4900308 (7.83%)  | 57025432 (91.14%)   | 643028 (1.03%)        | 92.17%        |



**Supplemental Table S2.** Correlation analysis of *ERF109*, *NAC35*, *NAC1*, *WRKY2D*, *NAC73*, *WRKY51*, *WRKY71*, *WRKY65* and *CAS1* expressions.

| var     | ERF109   | ERFTINY  | NAC73    | NAC35    | NAC1     | WRKY71   | WRKY2D   | WRKY51   | WRKY65   | CAS1 |
|---------|----------|----------|----------|----------|----------|----------|----------|----------|----------|------|
| ERF109  | 1        |          |          |          |          |          |          |          |          |      |
| ERFTINY | 0.274603 | 1        |          |          |          |          |          |          |          |      |
| NAC73   | 0.041235 | 0.918188 | 1        |          |          |          |          |          |          |      |
| NAC35   | -0.03186 | 0.85506  | 0.986481 | 1        |          |          |          |          |          |      |
| NAC1    | -0.06841 | 0.815317 | 0.975185 | 0.997047 | 1        |          |          |          |          |      |
| WRKY71  | -0.29467 | 0.730234 | 0.685307 | 0.606109 | 0.577815 | 1        |          |          |          |      |
| WRKY2D  | -0.10824 | 0.814557 | 0.976675 | 0.994571 | 0.998005 | 0.619157 | 1        |          |          |      |
| WRKY51  | 0.760358 | 0.721063 | 0.4163   | 0.291809 | 0.228071 | 0.366863 | 0.215379 | 1        |          |      |
| WRKY65  | 0.957618 | 0.433593 | 0.144246 | 0.060763 | 0.00751  | -0.11786 | -0.03018 | 0.877104 | 1        |      |
| CAS1    | 0.898006 | -0.15969 | -0.34682 | -0.40096 | -0.4166  | -0.59581 | -0.45283 | 0.453981 | 0.762316 | 1    |

Supplemental Table S3. Identification of the persulfidation peptide by LC-MS/MS.

| Sequence                     | # PSMs | #Proteins | #Protein | Protein        | Group | Modifications                               | ΔCn    | q-    | PEP         | IonScore | Exp Value   | Charge | MH+ [Da]   | ΔM    | RT    | # Missed  |
|------------------------------|--------|-----------|----------|----------------|-------|---------------------------------------------|--------|-------|-------------|----------|-------------|--------|------------|-------|-------|-----------|
|                              |        |           | Groups   | Accessions     |       |                                             |        | Value |             |          |             |        |            | [ppm] | [min] | Cleavages |
| EVAVSSGSAcTSASLEPSYVLR       | 2      | 1         | 1        | XP_004231957.1 |       | C10(Carbamidomethyl)                        | 0.0000 | 0     | 1.38851E-08 | 139      | 6.2361E-13  | 2      | 2270.09477 | -0.98 | 45.24 | 0         |
| EAAWAISNATSGGSHEQIK          | 2      | 2         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.000001672 | 101      | 3.80119E-09 | 2      | 1956.94267 | 0.81  | 25.78 | 0         |
| FDPDGEILEHLEAK               | 4      | 1         | 1        | XP_025885167.1 |       |                                             | 0.0000 | 0     | 0.0009044   | 98       | 5.07313E-09 | 2      | 1699.81316 | -2.52 | 42.51 | 0         |
| KLESLPSmVAGVWSNDNNmQLEATTQFR | 1      | 1         | 1        | XP_025886227.1 |       | M8(Oxidation);<br>M19(Oxidation)            | 0.0000 | 0     | 2.003E-07   | 98       | 6.24448E-09 | 3      | 3198.53098 | 3.56  | 48.75 | 1         |
| SFGVPPNTLAEYALNQFANK         | 2      | 1         | 1        | XP_004245632.1 |       |                                             | 0.0000 | 0     | 0.000004142 | 92       | 3.19508E-08 | 2      | 2181.09673 | -0.39 | 52.89 | 0         |
| SPPIEEVIQSGVVPR              | 1      | 1         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.0001039   | 88       | 4.1246E-08  | 2      | 1606.88005 | 0.04  | 48.57 | 0         |
| APGGcNNPcTVFK                | 1      | 2         | 1        | XP_004245632.1 |       | C5(Carbamidomethyl);<br>C9(Carbamidomethyl) | 0.0000 | 0     | 0.0003329   | 87       | 4.49212E-08 | 2      | 1421.63091 | 0.57  | 21.46 | 0         |
| FLSQPFFVAEVTGSPGK            | 2      | 2         | 1        | CAB65168.1     |       |                                             | 0.0000 | 0     | 0.00003409  | 82       | 2.82387E-07 | 2      | 1958.00725 | 0.68  | 53.69 | 0         |
| IINEPTAAAIAYGLDK             | 2      | 19        | 1        | AAA99920.1     |       |                                             | 0.0000 | 0     | 0.0004878   | 79       | 5.27642E-07 | 2      | 1659.89556 | 0.16  | 46.23 | 0         |
| EQAVWALGNVAGDSPR             | 2      | 1         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.00004063  | 78       | 8.0994E-07  | 2      | 1669.83220 | 1.71  | 41.44 | 0         |
| TKPHVNVGTIGHVDHGK            | 5      | 2         | 1        | XP_004247491.1 |       |                                             | 0.0000 | 0     | 0.00002116  | 73       | 2.16501E-06 | 2      | 1795.95708 | 0.46  | 11.52 | 0         |
| DLVLSNGALIPLLSQLNEHAK        | 2      | 1         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.00001185  | 70       | 0.000001575 | 2      | 2245.25810 | 1.31  | 53.70 | 0         |
| VDDVINVSGHR                  | 1      | 3         | 1        | XP_004242910.1 |       |                                             | 0.0000 | 0     | 0.007644    | 69       | 4.73122E-06 | 2      | 1210.61882 | 1.07  | 19.60 | 0         |
| HYAHVDcPGHADYVK              | 3      | 4         | 1        | XP_004247491.1 |       | C7(Carbamidomethyl)                         | 0.0000 | 0     | 0.0001835   | 69       | 2.92538E-06 | 2      | 1768.78520 | -0.48 | 12.01 | 0         |
| LVELLMHPSPSVLIPALR           | 2      | 1         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.005009    | 68       | 1.20014E-06 | 2      | 1985.16350 | 0.90  | 52.45 | 0         |
| FITcPScGHNIQLQHQR            | 2      | 1         | 1        | XP_025885167.1 |       | C4(Carbamidomethyl);<br>C7(Carbamidomethyl) | 0.0000 | 0     | 0.002031    | 66       | 9.9628E-06  | 3      | 2095.98337 | -3.80 | 20.99 | 0         |
| DDQTSTfTcPAGTNYR             | 1      | 1         | 1        | XP_004245632.1 |       | C9(Carbamidomethyl)                         | 0.0000 | 0     | 0.00155     | 66       | 3.97482E-06 | 2      | 1833.77507 | 2.29  | 24.96 | 0         |
| DVSGEGVQQALLK                | 1      | 4         | 1        | XP_004234968.1 |       |                                             | 0.0000 | 0     | 0.004133    | 65       | 1.44656E-05 | 2      | 1343.71868 | 1.55  | 32.10 | 0         |
| IQNAGTEVVEAK                 | 1      | 2         | 1        | XP_004235981.1 |       |                                             | 0.0000 | 0     | 0.003342    | 65       | 1.47195E-05 | 2      | 1258.66631 | 1.98  | 15.49 | 0         |
| TGcNFDASGK                   | 1      | 1         | 1        | XP_004245632.1 |       | C3(Carbamidomethyl)                         | 0.0000 | 0     | 0.01236     | 62       | 8.56918E-06 | 2      | 1056.44097 | -0.54 | 10.02 | 0         |
| VSIDNLNTNLPNAR               | 1      | 1         | 1        | XP_004234069.1 |       |                                             | 0.0000 | 0     | 0.002056    | 61       | 2.8646E-05  | 2      | 1540.80950 | 1.06  | 34.51 | 0         |
| LVELLmHPSPSVLIPALR           | 1      | 1         | 1        | XP_025886227.1 |       | M6(Oxidation)                               | 0.0000 | 0     | 0.001511    | 59       | 1.09275E-05 | 2      | 2001.15935 | 1.36  | 51.29 | 0         |
| VVIDHGAVPIFVK                | 2      | 1         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.008333    | 57       | 1.96351E-05 | 2      | 1393.81706 | -2.30 | 38.98 | 0         |
| GFEGGQMPLYR                  | 2      | 1         | 1        | XP_004238931.1 |       |                                             | 0.0000 | 0     | 0.02856     | 56       | 8.07612E-05 | 2      | 1254.60173 | 6.46  | 31.47 | 0         |
| EAFPGDVfYlhsR                | 4      | 1         | 1        | YP_009430460.1 |       |                                             | 0.0000 | 0     | 0.003132    | 55       | 0.000128772 | 3      | 1537.74387 | 0.27  | 43.20 | 0         |
| AEINQcQcPNELK                | 1      | 1         | 1        | XP_004245632.1 |       | C7(Carbamidomethyl)                         | 0.0000 | 0     | 0.01133     | 54       | 0.000124825 | 2      | 1443.68425 | -3.78 | 16.24 | 0         |
| ATLEDLCLR                    | 1      | 3         | 1        | XP_010315894.1 |       |                                             | 0.0000 | 0     | 0.1156      | 53       | 0.000272534 | 2      | 1045.54888 | -3.44 | 21.06 | 0         |
| LLASPSDDVR                   | 1      | 1         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.06501     | 50       | 0.000334432 | 2      | 1072.56413 | 0.71  | 19.89 | 0         |
| LPGVNKDGQVR                  | 1      | 1         | 1        | XP_025885167.1 |       |                                             | 0.0000 | 0     | 0.1036      | 50       | 0.000237827 | 2      | 1182.65827 | -0.63 | 13.08 | 1         |
| ISGLIYEETR                   | 1      | 5         | 1        | NP_001234121.1 |       |                                             | 0.0000 | 0     | 0.04232     | 50       | 0.00050524  | 2      | 1180.61919 | -1.43 | 28.62 | 0         |
| IVEIYGPESSGK                 | 1      | 1         | 1        | XP_004243994.1 |       |                                             | 0.0000 | 0     | 0.101       | 47       | 0.001036291 | 2      | 1278.65935 | 1.32  | 25.25 | 0         |
| EAcWTISNITAGNK               | 1      | 1         | 1        | XP_025886227.1 |       | C3(Carbamidomethyl)                         | 0.0000 | 0     | 0.004524    | 46       | 0.000928632 | 2      | 1564.74175 | -0.47 | 37.16 | 0         |
| THGQPASPTTLGK                | 1      | 1         | 1        | XP_004242661.1 |       |                                             | 0.0000 | 0     | 0.1036      | 45       | 0.001490529 | 2      | 1294.67143 | -2.79 | 10.35 | 0         |
| DITLGFVDLLRDDFVEQDR          | 1      | 12        | 1        | AID55406.1     |       |                                             | 0.0000 | 0     | 0.01383     | 39       | 0.006023736 | 3      | 2266.13687 | 0.77  | 54.98 | 1         |
| IQAVIEAGVcPR                 | 1      | 1         | 1        | XP_025886227.1 |       | C10(Carbamidomethyl)                        | 0.0000 | 0     | 0.1047      | 38       | 0.004257416 | 2      | 1312.70378 | -0.36 | 28.99 | 0         |
| NNPVLIGEPGVGK                | 2      | 5         | 1        | XP_004235966.1 |       |                                             | 0.0000 | 0     | 0.03399     | 34       | 0.011736413 | 2      | 1293.71587 | -0.26 | 29.24 | 0         |
| cPDAYSYPK                    | 1      | 1         | 1        | XP_004245632.1 |       | C1(Carbamidomethyl)                         | 0.0000 | 0     | 0.1156      | 34       | 0.005319464 | 2      | 1100.47173 | -0.02 | 15.84 | 0         |
| GKQPPPFEQVRPALSLQR           | 1      | 1         | 1        | XP_025886227.1 |       |                                             | 0.0000 | 0     | 0.01814     | 30       | 0.022802444 | 4      | 2119.17983 | 1.25  | 36.78 | 0         |

|                            |   |   |   |                |                       |        |   |         |    |             |   |            |          |       |   |
|----------------------------|---|---|---|----------------|-----------------------|--------|---|---------|----|-------------|---|------------|----------|-------|---|
| GAAELIPR                   | 1 | 1 | 1 | XP_004245089.1 |                       | 0.0000 | 0 | 0.08821 | 24 | 0.043836637 | 2 | 826.48021  | 2.46     | 23.73 | 0 |
| GLGYALSTTDELK              | 1 | 2 | 1 | XP_019068125.1 |                       | 0.0000 | 0 | 0.02195 | 23 | 0.224374323 | 2 | 1367.70842 | 2.25     | 35.50 | 0 |
| GVGPYTGVDGR                | 1 | 1 | 1 | XP_004236089.1 |                       | 0.0000 | 0 | 0.05451 | 23 | 0.223357398 | 2 | 1134.55669 | 2.48     | 18.78 | 0 |
| TFcIPHGGGGPGMGPIGVK        | 1 | 1 | 1 | XP_004245101.1 | C3(Carbamidomethyl)   | 0.0000 | 0 | 0.01497 | 19 | 0.576472164 | 3 | 1838.89066 | -7.33    | 38.07 | 0 |
| GILHDLPLPAGVK              | 1 | 1 | 1 | XP_025885167.1 |                       | 0.0000 | 0 | 0.1156  | 17 | 0.431960579 | 3 | 1386.81186 | 1.02     | 45.05 | 0 |
| IHPLIDDFILTIDGQNGlcYTHPQK  | 1 | 1 | 1 | XP_025885167.1 | C19(Carbamidomethyl)  | 0.0000 | 0 | 0.115   | 16 | 1.119581859 | 4 | 2908.45376 | -4.31    | 52.21 | 0 |
| KGlcCLCCKNNIDSLLYR         | 1 | 1 | 1 | XP_010313314.1 | C4(S)                 | 0.0000 | 0 | 0.01745 | 11 | 3.103013011 | 4 | 2262.04140 | 75628.01 | 34.91 | 2 |
| FYcPYKDCSamLVNDSDEIVR      | 1 | 1 | 1 | XP_004242195.1 | C3(S); M11(Oxidation) | 0.0000 | 0 | 0.06783 | 8  | 1.777149098 | 5 | 2573.08820 | 22163.27 | 6.66  | 1 |
| GGEVLVcADSccPIAVHISCMGCPAR | 3 | 1 | 1 | XP_004243220.1 | C7(S); C12(S)         | 0.0000 | 0 | 0.01063 | 1  | 6.2994365   | 5 | 2826.17048 | 60537.62 | 7.57  | 0 |
| SScmICLPCRHLSSCK           | 1 | 1 | 1 | XP_004231045.1 | C3(S); M4(Oxidation)  | 0.0000 | 0 | 0.01116 | 0  | 5.819048965 | 4 | 1986.79787 | 86092.14 | 18.31 | 1 |

**Supplemental Table S4.** Primers used in this study for RT-qPCR analysis.

| Primers                    | Sequence                     | Gene ID          |
|----------------------------|------------------------------|------------------|
| <i>SlTubulin</i> (forward) | TAGAGCCTGGTACGATGGATAG       | Solyc08g006890   |
| <i>SlTubulin</i> (reverse) | CAACTCAGCGCCTTCAGTATAA       |                  |
| <i>SISGR1</i> (forward)    | GGCTATCTCCCAAACCATCAA        | Solyc08g080090   |
| <i>SISGR1</i> (reverse)    | ACTCTGCAACAACCTTCATCTCT      |                  |
| <i>SIPPH</i> (forward)     | CAGATGTCGTGGATGGGAAA         | Solyc01g088090   |
| <i>SIPPH</i> (reverse)     | GCCTTTAGCTTCACCTTGGTAAAC     |                  |
| <i>SIPAO</i> (forward)     | CCTCATCGTCTTGCTCCTTTAT       | Solyc11g066440   |
| <i>SIPAO</i> (reverse)     | GCAGCTTGAGGTATCCTTGT         |                  |
| <i>SINYC1</i> (forward)    | GATCAGGGAAGAGCACTGTATG       | Solyc07g024000   |
| <i>SINYC1</i> (reverse)    | GAGAACACAGACACCCAAAGTAT      |                  |
| <i>SIPSY1</i> (forward)    | GAAGATGCCAGAAGAGGAAGAG       | Solyc03g031860   |
| <i>SIPSY1</i> (reverse)    | GGTCACCCTTCCAGCAAATA         |                  |
| <i>SIPDS</i> (forward)     | CAAGACCAGAGCTGGACAATAC       | Solyc03g123760   |
| <i>SIPDS</i> (reverse)     | CAAACCTGCACCAGCAATAAC        |                  |
| <i>SIZDS</i> (forward)     | GATTGGTTCCCTCAGAAGTGGAG      | Solyc01g097810   |
| <i>SIZDS</i> (reverse)     | CCAGCCATTGTAGCGTAGTT         |                  |
| <i>SIERFTINY</i> (forward) | GGAGTTAGGATGAGAAGTTGGG       | LOC101259323     |
| <i>SIERFTINY</i> (reverse) | TCCTTTTATACTCACCGCTGC        |                  |
| <i>SIERF109</i> (forward)  | GGAGTGAGACAGAGACCATG         | LOC101246484     |
| <i>SIERF109</i> (reverse)  | ATGTTCCAAGCCATACTCTAGC       |                  |
| <i>SINAC73</i> (forward)   | AAGGATGGTCAAGTGCGTC          | LOC101264084     |
| <i>SINAC73</i> (reverse)   | ACCTGTCTTGTGCCATCTTG         |                  |
| <i>SINAC35</i> (forward)   | GAAATGAAGCACGTACACAGC        | LOC101267943     |
| <i>SINAC35</i> (reverse)   | CTTCATCCTTGCTGATTGACG        |                  |
| <i>SINAC1</i> (forward)    | TCGATGACTAGAATGCACACG        | Solyc04g009440.2 |
| <i>SINAC1</i> (reverse)    | TCAAGGGTGTTTCTGTGGTC         |                  |
| <i>SIWRKY2D</i> (forward)  | GTGTATTCTCCAACTCCGATCC       | Solyc02g093050.2 |
| <i>SIWRKY2D</i> (reverse)  | TCTGTTTCCCCTGTAAACGAC        |                  |
| <i>SIWRKY51</i> (forward)  | AGAGTTGGAGGTGTTGGATG         | LOC101258361     |
| <i>SIWRKY51</i> (reverse)  | ATCTTTATTGTCCCTTTCTACTCTTTTC |                  |
| <i>SIWRKY65</i> (forward)  | GAACACAATCACCAAATCCCG        | LOC101267623     |
| <i>SIWRKY65</i> (reverse)  | GTAGAGGTAGTGGGTGAACTG        |                  |
| <i>SIWRKY71</i> (forward)  | TCATCTACCAGCTACTCTCCG        | LOC101264783     |
| <i>SIWRKY71</i> (reverse)  | GGCATGTCCATAAAATTGAGGC       |                  |
| <i>SICAS1</i> (forward)    | CCAACTAAAGGAATGGGAGGAA       | LOC101253940     |
| <i>SICAS1</i> (reverse)    | GAGTATTGGCAGGATTGGAGAA       |                  |

**Supplemental Table S5.** Primers used in this study for constructing experiment-related vectors.

| Primers                                 | Sequences (5'-3')                                       | Gene ID      |
|-----------------------------------------|---------------------------------------------------------|--------------|
| <i>SIWRKY71</i> -CRISPR/Cas9-target 1-F | GTCAAGTGAAGAAGCGTGTAGAG                                 | LOC101264783 |
| <i>SIWRKY71</i> -CRISPR/Cas9-target 1-R | AAACCTCTACACGCTTCTTCACT                                 |              |
| <i>SIWRKY71</i> -CRISPR/Cas9-target 2-F | GTCACCTCTCCGCGGTAGCGTCGCA                               |              |
| <i>SIWRKY71</i> -CRISPR/Cas9-target 2-R | AAACTGCGACGCTACCGCGGAGAG                                |              |
| <i>SIWRKY71</i> -pBI121-F               | GAGAACACGGGGGACTCTAGAATGGGTGATGAACTAAG<br>AGATTTGTACT   |              |
| <i>SIWRKY71</i> -pBI121-R               | CTTGTAATCGGTACCCTCGAGTGGTTCTTGTTTAAGATTA<br>AACATAGAAGG |              |
| <i>SIWRKY71</i> -pCold-F                | CTCGGTACCCTCGAGGGATCCATGGGTGATGAACTAAGA<br>GATTTGTACT   |              |
| <i>SIWRKY71</i> -pCold-R                | AGCAGAGATTACCTATCTAGATCATGGTTCTTGTTTAAGA<br>TTAAACATAG  |              |
| <i>SIWRKY71</i> -pSAK277-F              | ACTAGTGGATCCAAAGAATTCATGGGTGATGAACTAAGA<br>GATTTGTACT   |              |
| <i>SIWRKY71</i> -pSAK277-R              | TCATTAAAGCAGGACTCTAGATCATGGTTCTTGTTTAAGA<br>TTAAACATAG  |              |
| <i>SIWRKY71</i> -vp64-F                 | ACCTGGACATGCTCGGATCAATGGGTGATGAACTAAGAG<br>ATTG         |              |
| <i>SIWRKY71</i> -vp64-R                 | GAAGTACTCTCGAGAAGCTTTCATGGTTCTTGTTTAAGAT<br>TAAACA      |              |
| <i>SIWRKY71</i> -F1-vp64-F              | ACCTGGACATGCTCGGATCAATGGGTGATGAACTAAGAG<br>ATTG         |              |
| <i>SIWRKY71</i> -F1-vp64-R              | GAAGTACTCTCGAGAAGCTTTGTTTTCTCAGGCTTTCTCT<br>GTC         |              |
| <i>SIWRKY71</i> -F2-vp64-F              | ACCTGGACATGCTCGGATCAGTGGAGTTGTCAAAGGTTA<br>C            |              |
| <i>SIWRKY71</i> -F2-vp64-R              | GAAGTACTCTCGAGAAGCTTTCATGGTTCTTGTTTAAGAT<br>TAAACA      |              |
| <i>SIWRKY71</i> -TRV2-F                 | CGTGAGCTCGGTACCGGATCCATGGGTGATGAACTAAGA<br>GATTTGTACT   |              |
| <i>SIWRKY71</i> -TRV2-R                 | GTGAGTAAGGTTACCGAATTCCACCATCTTCTAATGTTTC<br>ATCCTTG     |              |
| <i>SIWRKY71</i> -pGBKT7-F               | TCAGAGGAGGACCTGCATATGATGGGTGATGAACTAAGA<br>GATTTGTACT   |              |
| <i>SIWRKY71</i> -pGBKT7-R               | ACGATTCATCTGCAGCTCGAGTGGTTCTTGTTTAAGATTA<br>AACATAGAAGG |              |
| <i>SIWRKY71</i> -pGBKT7-S1-F            | TCAGAGGAGGACCTGCATATGATGGGTGATGAACTAAGA<br>GATTTGTACT   |              |
| <i>SIWRKY71</i> -pGBKT7-S1-R            | CTAGTTATGCGGCCGCTGCAGATGATCAACTTCACTCTTT<br>GTCATGAA    |              |

|                                       |                                                               |
|---------------------------------------|---------------------------------------------------------------|
| <i>SIWRKY71</i> -pGBKT7-S2-F          | TCAGAGGAGGACCTGCATATGCTTGAAGATGGATATAGAT<br>GGAGGAA           |
| <i>SIWRKY71</i> -pGBKT7-S2-R          | CTAGTTATGCGGCCGCTGCAGTAGATGATGGTTGTGTTGA<br>CCTTCA            |
| <i>SIWRKY71</i> -pGBKT7-S3-F          | TCAGAGGAGGACCTGCATATGCCAGCTACTCTCCGCGGT<br>AG                 |
| <i>SIWRKY71</i> -pGBKT7-S3-R          | CTAGTTATGCGGCCGCTGCAGTGGTTCTTGTTTAAGATTA<br>AACATAGAAGG       |
| <i>SIWRKY71</i> -pGADT7-F             | GCCATGGAGGCCAGTGAATTCATGGGTGATGAACTAAGA<br>GATTTGTACT         |
| <i>SIWRKY71</i> -pGADT7-F             | ACGATTCATCTGCAGCTCGAGTGGTTCTTGTTTAAGATTA<br>AACATAGAAGG       |
| <i>SIWRKY71</i> -nLuc-F               | CGAGCTCGGTACCCGGGATCCATGGGTGATGAACTAAGA<br>GA                 |
| <i>SIWRKY71</i> -nLuc-R               | CGCGTACGAGATCTGGTCGACTGGTTCTTGTTTAAGATTA<br>AACAT             |
| <i>SIWRKY71</i> -cLuc-F               | CCGGGGCGGTACCCGGGATCCATGGGTGATGAACTAAGA<br>GATT               |
| <i>SIWRKY71</i> -cLuc-R               | ACGAAAGCTCTGCAGGTCGACTCATGGTTCTTGTTTAAG<br>ATTAAACA           |
| <i>SlCAS1</i> -CRISPR/Cas9-target 1-F | GTCACATCAATGGCGTCGTTGGTG                                      |
| <i>SlCAS1</i> -CRISPR/Cas9-target 1-R | AAACCACCAACGACGCCATTGATG                                      |
| <i>SlCAS1</i> -CRISPR/Cas9-target 2-F | GTCAGACTACTGGCCCTGAAATC                                       |
| <i>SlCAS1</i> -CRISPR/Cas9-target 2-F | AAACGATTTTCAGGGCCAGTAGTC                                      |
| <i>SlCAS1</i> -pBI121-F               | ACGGGGGACTCTAGAGGATCCATGGCGTCGTTGGTGAGG<br>A                  |
| <i>SlCAS1</i> -pBI121-R               | CGATCGGGGAAATTCGAGCTCCTAATCAACTGATACAGG<br>TTGCATGTT          |
| <i>SlCAS1</i> -pGreenII0800LUC-F      | GTCGACGGTATCGATAAGCTTAATGAGTGTGATGATAATT<br>TAAGAACTACAA      |
| <i>SlCAS1</i> -pGreenII0800LUC-R      | CGCTCTAGAACTAGTGGATCCTGATGAAATTTATTTGAAG<br>CCTGTAA           |
| <i>SlCAS1</i> -pAbAi-S1-F             | AAATGATGAATTGAAAAGCTTCTACAATAATTTGTAGTAG<br>TACTCCTTATATTTTAA |
| <i>SlCAS1</i> -pAbAi-S1-R             | TACAGAGCACATGCCTCGAGAGTTAGGAAGGTGGACCT<br>GTGATG              |
| <i>SlCAS1</i> -pAbAi-S2-F             | AAATGATGAATTGAAAAGCTTGTTATACCCACTTTGAGCC<br>ATCACA            |
| <i>SlCAS1</i> -pAbAi-S2-R             | ATACAGAGCACATGCCTCGAGCACATATAGAGATAGTAA<br>CTGTTTGAAGTTAAA    |
| <i>SlCAS1</i> -pAbAi-S3-F             | AAATGATGAATTGAAAAGCTTGTTTGGATTGAATTTTCAC<br>TTTTGA            |
| <i>SlCAS1</i> -pAbAi-S3-R             | ATACAGAGCACATGCCTCGAGTCTACTTGTTTTCCCGGA<br>AAACA              |

LOC101253940

|                              |                                                                |              |
|------------------------------|----------------------------------------------------------------|--------------|
| <i>SICAS1</i> -TRV2-F        | GTGAGTAAGGTTACCGAATTCCATGCAACCTGTATCAGTT<br>GATTAGTG           |              |
| <i>SICAS1</i> -TRV2-R        | CGTGAGCTCGGTACCGGATCCGCAAAGCATTTTATGGTAT<br>GATAAATC           |              |
| <i>SIBRG3</i> -pMAL-F        | AGGATTTCAGAATTCGGATCCATGGCTATTCAAGCGC<br>AATT                  |              |
| <i>SIBRG3</i> -pMAL-R        | ACGACGGCCAGTGCCAAGCTTTCAAAACATGGCTTC<br>AATGGC                 |              |
| <i>SIBRG3</i> -mutant-pMAL-F | GATCTTCGGCGATGATTTGCTTACCCGCGAGACAT<br>CT                      |              |
| <i>SIBRG3</i> -mutant-pMAL-R | AGATGTCTCGCGGGTAAGCAAATCATCGCCGAAG<br>ATC                      |              |
| <i>SIBRG3</i> -pGADT7-F      | GCCATGGAGGCCAGTGAATTCATGGCTATTCAAG<br>CGCAATTG                 |              |
| <i>SIBRG3</i> -pGADT7-R      | ACGATTCATCTGCAGCTCGAGTCAAAACATGGCT<br>TCAATGGC                 | LOC101267168 |
| <i>SIBGR3</i> -nLuc-F        | CGAGCTCGGTACCCGGGATCCATGGCTATTCAAG<br>CGCAATTG                 |              |
| <i>SIBGR3</i> -nLuc-R        | CGCGTACGAGATCTGGTCGACAAACATGGCTTCA<br>ATGGCAGCT                |              |
| <i>SIBGR3</i> -cLuc-F        | CCGGGGCGGTACCCGGGATCCATGGCTATTCAAG<br>CGCAATTG                 |              |
| <i>SIBGR3</i> -cLuc-R        | ACGAAAGCTCTGCAGGTCGACTCAAAACATGGCT<br>TCAATGGCA                |              |
| <i>SIRNF-144b</i> -nLuc-F    | CGAGCTCGGTACCCGGGATCCATGGCTATTCAAGCGCAA<br>TTG                 |              |
| <i>SIRNF-144b</i> -nLuc-R    | CGCGTACGAGATCTGGTCGACAAACATGGCTTCAATGGC<br>AGCT                |              |
| <i>SIRNF-144b</i> -cLuc-F    | CCGGGGCGGTACCCGGGATCCATGGCTATTCAAGCGCAA<br>TTG                 | LOC101265447 |
| <i>SIRNF-144b</i> -cLuc-R    | ACGAAAGCTCTGCAGGTCGACTCAAAACATGGCTTCAAT<br>GGCA                |              |
| <i>SIERFTINY</i> -pSAK277-F  | ACTAGTGGATCCAAAGAATTCATGTCTACTACACATCGCT<br>ATAAAAACG          |              |
| <i>SIERFTINY</i> -pSAK277-R  | TCATTAAAGCAGGACTCTAGATAATATTAACATATTTCTT<br>CTATAATCAAATGTATAA | LOC101259323 |
| <i>SIERF109</i> -pSAK277-F   | ACTAGTGGATCCAAAGAATTCATGCAGAGGTCAACAAAA<br>CTTAGTG             |              |
| <i>SIERF109</i> -pSAK277-R   | TCATTAAAGCAGGACTCTAGACTAGAAGCTATGAGCAAT<br>ATAGCCATT           | LOC101246484 |
| <i>SINAC73</i> -pSAK277-F    | ACTAGTGGATCCAAAGAATTCATGGCATGGTGCAGTCAA<br>TCA                 |              |
| <i>SINAC73</i> -pSAK277-R    | TCATTAAAGCAGGACTCTAGATCACTTCTCTTTGCTTGCT<br>TCTGA              | LOC101264084 |

|                            |                                                           |                  |
|----------------------------|-----------------------------------------------------------|------------------|
| <i>SINAC35</i> -pSAK277-F  | ACTAGTGGATCCAAAGAATTCATGGAAGATCTACCAGGG<br>TTCCG          | LOC101267943     |
| <i>SINAC35</i> -pSAK277-R  | TCATTAAAGCAGGACTCTAGACTAAAAGTTATGCACAAA<br>CGCGG          |                  |
| <i>SINAC1</i> -pSAK277-F   | ACTAGTGGATCCAAAGAATTCATGAACAAAGGAGCAAA<br>CGGAA           | Solyc04g009440.2 |
| <i>SINAC1</i> -pSAK277-R   | TCATTAAAGCAGGACTCTAGATTAGTAAGGTTTTTGCATG<br>TATAGGAAC     |                  |
| <i>SIWRKY2D</i> -pSAK277-F | ACTAGTGGATCCAAAGAATTCATGGCTGTGGAGCTAATG<br>ATGG           | Solyc02g093050.2 |
| <i>SIWRKY2D</i> -pSAK277-R | TCATTAAAGCAGGACTCTAGATTAAGAAGACTCTAAAAT<br>GAGACTACTTGTTT |                  |
| <i>SIWRKY51</i> -pSAK277-F | ACTAGTGGATCCAAAGAATTCATGGAAAATTTCCCTATA<br>GCTCATC        | LOC101258361     |
| <i>SIWRKY51</i> -pSAK277-R | TCATTAAAGCAGGACTCTAGACTAAAGGTGAAGATTGTG<br>AAGGGC         |                  |
| <i>SIWRKY65</i> -pSAK277-F | ACTAGTGGATCCAAAGAATTCATGGAAGATAGTTCATAC<br>AAAAATCTATTTT  | LOC101267623     |
| <i>SIWRKY65</i> -pSAK277-R | TCATTAAAGCAGGACTCTAGATTATCCTGTACCTCCGCAG<br>CA            |                  |



**Supplemental Table S6.** Primers used for PCR amplification of fragments containing the target sites of gRNA of CRISPR/Cas9-*WRKY71* and CRISPR/Cas9-*CASI*.

| Gene            | Forward primer (5'-3')      | Reverse primer (5'-3')      |
|-----------------|-----------------------------|-----------------------------|
| <i>SIWRKY71</i> | F: GTTGATCATCTTGAAGATGGATAT | R: ATTTGCAAAAACATTGGCATGTCC |
| <i>SICASI</i>   | F1: GAGATAACAAGCTCGGCC      | R1: GAGATAACAAGCTCGGCC      |
|                 | F2: AGCTCCTTCTGAGTCCAG      | R2: CCCCAT TGCCAGTTATTTC    |